# Chapter 9 — Errors, Exceptions and Files

Programs meet bad input, missing files, full disks and broken networks. Python reports these situations with **exceptions** — objects that interrupt normal flow and travel up the call stack (the pile of function calls currently in progress) until some code handles them. Most of the time the thing that fails is I/O (input/output: reading and writing files, networks, devices), so this chapter pairs error handling with working on files: reading and writing text and bytes, paths, CSV, JSON, pickle (three data file formats, §9.11), temporary files and whole-folder operations.

Scripts are in `code/`; every file operation happens inside a temporary directory that is removed afterwards. Output is real (Python 3.14; code is 3.12-compatible).

**What you'll learn**

| # | Topic | Script |
|---|---|---|
| 9.1 | What exceptions are, tracebacks, the built-in hierarchy | `01_exceptions.py` |
| 9.2 | `try` / `except` / `else` / `finally` | `01_exceptions.py` |
| 9.3 | Raising, re-raising, custom exception classes | `01_exceptions.py` |
| 9.4 | Exception chaining (`raise … from …`), `add_note` | `01_exceptions.py` |
| 9.5 | EAFP vs LBYL | `01_exceptions.py` |
| 9.6 | `ExceptionGroup` and `except*` (3.11+) | `01_exceptions.py` |
| 9.7 | `assert` | `01_exceptions.py` |
| 9.8 | Context managers and `with` | `02_context_managers.py` |
| 9.9 | File I/O: modes, text vs binary, encodings | `03_files_pathlib.py` |
| 9.10 | `pathlib`, `shutil`, `tempfile` | `03_files_pathlib.py` |
| 9.11 | `csv`, `json`, `pickle` (and why pickle is unsafe) | `04_csv_json_pickle.py` |

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## 9.1 Exceptions and the hierarchy

### What it is

An **exception** is an object describing an error, **raised** (thrown) at the point of failure. Raising stops the current code immediately; Python then unwinds the call stack — leaving each function in turn — looking for a matching `except` handler. If none is found, the program stops and prints a **traceback**: the chain of calls that led to the error (read it **bottom-up**: the last line is the exception type and message, the line above is where it happened).

```
Traceback (most recent call last):
  File "app.py", line 8, in <module>
    total = parse("12a")
  File "app.py", line 3, in parse
    return int(text)
ValueError: invalid literal for int() with base 10: '12a'
```

Two kinds of problems are distinct:

- **Syntax errors** are found when Python *compiles* the file (translates it to bytecode before running) (`SyntaxError`, `IndentationError`) — nothing runs.

- **Exceptions** happen while the code *runs* (`ZeroDivisionError`, `KeyError`, …) and can be handled.

Analogy: an exception is a fire alarm. Whoever is nearest and equipped (a matching `except`) deals with it; if nobody does, the building (program) is evacuated.

### How it works — the class hierarchy

Every exception is an instance (an object made from a class) of a class deriving from (being a subclass of) **`BaseException`**. Handlers match by class *and all its subclasses*, so the tree decides what an `except` catches.

```
== part of the built-in hierarchy ==
  BaseException
      BaseExceptionGroup
      Exception
      GeneratorExit
      KeyboardInterrupt
      SystemExit
  ...under Exception (selection):
  LookupError
      IndexError
      KeyError
  ArithmeticError
      FloatingPointError
      OverflowError
      ZeroDivisionError
  OSError
      FileNotFoundError
      IsADirectoryError
      PermissionError
      TimeoutError
  issubclass(ModuleNotFoundError, ImportError): True
```

(The script walks `__subclasses__()` of the real classes; only a selection is printed.)

| Class | Raised when | Example |
|---|---|---|
| `SystemExit`, `KeyboardInterrupt`, `GeneratorExit` | Program exit, Ctrl-C, generator (a function that yields values one at a time) closing | Deliberately **not** under `Exception` so `except Exception` doesn't swallow (catch and hide) them |
| `ValueError` | Right type, bad value | `int("abc")` |
| `TypeError` | Wrong type / wrong call | `"a" + 1`, too many arguments |
| `KeyError` / `IndexError` (both `LookupError`) | Missing key / index out of range | `{}["x"]`, `[][0]` |
| `AttributeError` | Missing attribute | `None.upper()` |
| `NameError` (→ `UnboundLocalError`) | Unknown name | typo in a variable |
| `ZeroDivisionError` (`ArithmeticError`) | Division by zero | `1 / 0` |
| `OSError` family | OS-level failures | `FileNotFoundError`, `PermissionError`, `IsADirectoryError`, `TimeoutError`, `ConnectionError` |
| `ImportError` (→ `ModuleNotFoundError`) | Import failures | Chapter 8 |
| `StopIteration` | Iterator exhausted | used by `for` internally |
| `RecursionError` (`RuntimeError`) | Too-deep recursion | Chapter 7 |
| `NotImplementedError` (`RuntimeError`) | Abstract method (declared by a base class for subclasses to fill in) not overridden | |
| `AssertionError` | `assert` failed | 9.7 |
| `UnicodeDecodeError` (→ `ValueError`) | Bytes aren't valid in the codec (the encoder/decoder for a text encoding) | 9.9 |

## In the interview

**30-second answer:** "An exception is an object raised on error; it unwinds the stack until an `except` clause whose class matches (including subclasses) handles it, otherwise the program ends with a traceback. All exceptions derive from `BaseException`; application errors derive from `Exception`, while `SystemExit`, `KeyboardInterrupt` and `GeneratorExit` sit directly under `BaseException` so generic handlers don't catch them."

- *Why is `KeyboardInterrupt` not an `Exception`?* → So `except Exception:` doesn't stop Ctrl-C from working.
- *How do you read a traceback?* → Bottom line = type and message; the frames above show the call path, most recent last.

## Self-check

1. Common parent of `KeyError` and `IndexError`? → `LookupError`.
2. Is `SyntaxError` catchable in the file that contains it? → No — the file never compiles, so nothing in it runs.

3. What is `FileNotFoundError` a subclass of? → `OSError`.

### TL;DR

Exceptions are objects that unwind the stack until matched by class. Know the tree: `BaseException` → `Exception` → your errors.

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## 9.2 `try` / `except` / `else` / `finally`

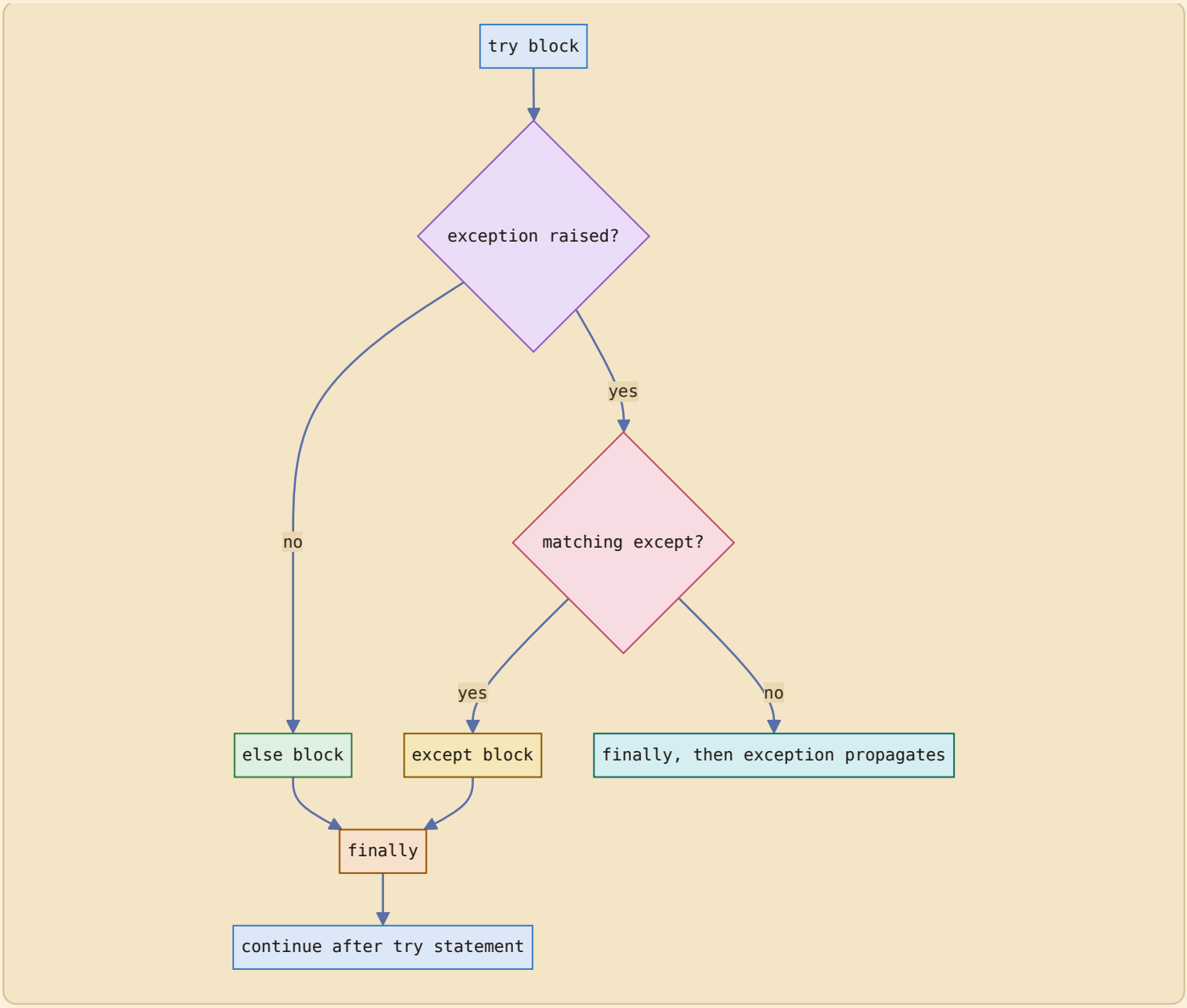
### What it is

The statement for handling exceptions:

```
try:
    risky()                 # code that might fail
except SomeError as exc:    # runs if SomeError (or subclass) was raised in try
    handle(exc)
else:                       # runs only if try raised NOTHING
    use_result()
finally:                    # runs ALWAYS: success, handled, unhandled, return, break
    clean_up()
```

### How it works

1. Run the `try` block.
2. If an exception occurs, skip the rest of `try` and check `except` clauses **top to bottom**; the **first** matching one runs. `except (A, B) as e:` catches either.
3. If nothing was raised, run `else`.
4. Run `finally` in every case — even if an exception is propagating (still travelling up the call stack), or the block `return`s.
5. If no clause matched, the exception continues up the stack (after `finally`).




```python
def parse_age(text):
    try:
        age = int(text)
    except ValueError:
        print(f"  except: {text!r} is not a number")
        return None
    else:
        print(f"  else:   parsed {age}")
        return age
    finally:
        print("  finally: always runs")
```

```
== try / except / else / finally ==
  else:   parsed 42
  finally: always runs
  except: 'forty' is not a number
  finally: always runs
```

`finally` ran even though both branches `return`ed.

**Why `else`?** It keeps the `try` block tiny — only the line that can raise the error you're handling. Code in `else` isn't protected, so a bug there won't be silently caught by the wrong handler.

## Trade-offs / rules of thumb

| Do | Don't |
|---|---|
| Catch the **most specific** exception you can handle | `except:` (bare) — also catches `KeyboardInterrupt` and `SystemExit` |
| Order handlers specific → general | Put `except Exception` first (it would shadow the rest) |
| Keep `try` blocks small | Wrap 50 lines in one `try` |
| Handle where you can *do something* (retry, default, message) | Catch just to `pass` (silently hiding bugs) |
| Use `finally` / `with` for cleanup | `return` inside `finally` — it swallows exceptions (Python 3.14 emits a `SyntaxWarning`, PEP 765) |

## When not to use it

Don't use exceptions for ordinary control flow when a simple check or API already exists (`dict.get`, `str.isdigit` for trusted formats). And don't catch errors you can't handle — let them reach a top-level handler that logs them.

## Common mistakes

- `except Exception as e: pass` — the bug disappears and resurfaces far away.
- Referencing `e` after the `except` block — Python **deletes** the `as` name at the end of the handler (to break reference cycles — objects that refer to each other in a loop and so can't be freed promptly), so it's a `NameError` later. Save it to another variable if needed.
- Catching `Exception` around code that includes your own bugs (`TypeError` from a typo) and reporting "invalid input".

## Real-world examples

- **Logging instead of hiding**: at the top level, `logging.exception("msg")` (inside an `except`) logs the message *plus* the full traceback; the `traceback` module (`traceback.print_exc()`, `traceback.format_exc()`) gives you the same text as a string or on stderr (the separate error-output stream). See Chapter 14.
- **Warnings are not exceptions**: `warnings.warn("old_api() is deprecated", DeprecationWarning)` prints a message but lets the program continue. Libraries use this for deprecations (features scheduled for removal); `python -W error` (or pytest's `-W error`) turns warnings into exceptions so you catch them in CI (continuous integration: automated checks on every code change).
- Web frameworks (Django, Flask, FastAPI) wrap every request in a top-level handler that turns uncaught exceptions into a 500 response (the HTTP "internal server error" status; HTTP basics are in System Design Fundamentals) and a log entry, so one bad request doesn't crash the server.
- `requests`: `resp.raise_for_status()` raises `requests.HTTPError` for 4xx/5xx (HTTP error statuses: 4xx = the client's fault, 5xx = the server's); all its errors derive from `requests.RequestException`, so callers can catch one base class.

## In the interview

> **30-second answer:** "`try` runs risky code; the first matching `except` handles; `else` runs only if nothing was raised, which keeps the `try` minimal; `finally` always runs for cleanup. Catch specific exceptions, order specific to general, never a bare `except`."

- *When does `else` run?* → Only when `try` completed without an exception.
- *Does `finally` run if `try` returns?* → Yes, just before the function actually returns.

- *Bare `except:` vs `except Exception:`?* → Bare also catches `BaseException` subclasses like `KeyboardInterrupt` / `SystemExit`.

### Self-check

1. Two handlers `except Exception` then `except ValueError` — which runs for `int("x")`? → The first (`Exception`); the second is dead code.
2. What's the point of `else`? → Separate "code that may raise the handled error" from "code that runs on success".
3. Is `e` defined after `except ValueError as e:` finishes? → No.

### TL;DR

`try` small, `except` specific (first match wins), `else` on success, `finally` always.

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## 9.3 Raising, re-raising and custom exceptions

### What it is

- **`raise SomeError("message")`** signals an error yourself — typically when inputs violate your function's contract (the promises it makes about valid inputs and outputs).
- **Re-raising**: inside an `except`, a bare **`raise`** re-throws the current exception with its original traceback, after you've logged or cleaned up.
- **Custom exceptions** are classes you define (subclassing `Exception`) so callers can catch *your* errors precisely.

### How it works

1. Pick the right built-in when one fits: bad value → `ValueError`, wrong type → `TypeError`, missing key-like thing → `KeyError` / `LookupError`, unsupported operation → `NotImplementedError`.
2. For domain errors (errors that mean something in your problem area, such as payments), define a **base class** for your library/module and specific subclasses. Callers can catch all your errors (`except PaymentError`) or one (`except CardDeclined`).
3. Name them with an `Error` suffix (PEP 8). Add attributes for data the handler needs.

```python
class PaymentError(Exception):
    """Base class for this module's errors."""

class CardDeclined(PaymentError):
    def __init__(self, card_last4, reason):
        super().__init__(f"card ending {card_last4} declined: {reason}")
        self.card_last4 = card_last4
        self.reason = reason

class InsufficientFunds(PaymentError):
    pass

def charge(balance, amount):
    if amount <= 0:
        raise ValueError(f"amount must be positive, got {amount}")
    if amount > balance:
        raise InsufficientFunds(f"need {amount}, have {balance}")
    return balance - amount

def reraise_after_logging(x):
    try:
        return 1 / x
    except ZeroDivisionError:
        print("  logging the error, then re-raising")
        raise                           # bare raise keeps the original traceback
```

```
== raising and catching custom exceptions ==
  charge(100, 30) -> 70
  InsufficientFunds: need 500, have 100
  ValueError: amount must be positive, got -5
  caught CardDeclined, attrs: last4=4242 reason=expired

== re-raising ==
  logging the error, then re-raising
  caller still sees: ZeroDivisionError('division by zero')
```

Pass the message to `super().__init__` (the parent class's initialiser) so `str(exc)` and tracebacks show it.

## Trade-offs

Too few custom exceptions → callers parse message strings to distinguish errors (fragile). Too many → a zoo nobody remembers. A module base class plus a handful of meaningful subclasses is the usual sweet spot. Many libraries also inherit from a matching built-in too (`class InvalidAmount(PaymentError, ValueError)`) so generic `except ValueError` code keeps working.

## Common mistakes

- Subclassing `BaseException` instead of `Exception` — escapes `except Exception` handlers.
- `raise exc` instead of bare `raise` inside the handler — works, but adds the current line to the traceback; bare `raise` is the idiom.
- Raising strings (`raise "error"`) — a `TypeError` in Python 3; only exception instances/classes can be raised.
- Returning error codes (`return -1`) instead of raising — callers forget to check.

## Real-world examples

- `requests.RequestException` → `ConnectionError`, `Timeout`, `HTTPError`…

- Django: `ObjectDoesNotExist` base with per-model `Model.DoesNotExist` subclasses; `ValidationError` for form/model validation.
- `json.JSONDecodeError` subclasses `ValueError`, so old code catching `ValueError` still works.

### In the interview

> **30-second answer:** "Raise built-ins like `ValueError` / `TypeError` for contract violations; for domain errors create a base exception subclassing `Exception` plus specific subclasses with useful attributes. Inside a handler, bare `raise` re-throws with the original traceback."

- *Why a base exception class per library?* → Callers can catch everything from your library with one clause.
- *`raise` vs `raise e`?* → Bare `raise` re-raises the active exception unchanged; `raise e` re-raises the object from the current line.

### Self-check

1. Which class should custom exceptions subclass? → `Exception` (or a subclass).
2. What does a bare `raise` outside any handler do? → `RuntimeError: No active exception to reraise`.
3. How do you attach data to a custom exception? → Set attributes in `__init__`.

### TL;DR

`raise` built-ins for contract errors; custom hierarchy (base + subclasses, `...Error`) for domain errors; bare `raise` to re-throw.

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## 9.4 Exception chaining and notes

### What it is

When you catch a low-level error and raise a higher-level one, Python remembers the link:

| Syntax | Stored in | Traceback says |
|---|---|---|
| `raise New(...)` inside an `except` (implicit) | `new.__context__` | "During handling of the above exception, another exception occurred" |
| `raise New(...) from orig` (explicit) | `new.__cause__` | "The above exception was the direct cause of the following exception" |
| `raise New(...) from None` | suppresses the context | Only the new exception is shown |

**`exc.add_note("…")`** (3.11+) attaches extra context lines that appear under the message in the traceback — handy for "which row/file/request was this?".

#### Why it matters

Translating errors keeps your API (the interface your code offers its callers) clean (callers catch `ConfigError`, not whatever `KeyError` / `ValueError` your internals produced) *without* losing the root cause (the original underlying failure) needed for debugging.

## How it works

```
class ConfigError(Exception):
    pass

def load_port(cfg):
    try:
        return int(cfg["port"])
    except (KeyError, ValueError) as exc:
        raise ConfigError("invalid 'port' setting") from exc
```

```
== exception chaining: raise ... from ... ==
  port = 8080
  ConfigError: invalid 'port' setting; __cause__ = KeyError('port')
  ConfigError: invalid 'port' setting; __cause__ = ValueError("invalid literal for int() with base 10:
'eighty'")
  implicit: __context__ = KeyError('missing')
  formatted traceback of an explicit chain (last lines):
    | Traceback (most recent call last):
    | KeyError: 'port'
    |
    | The above exception was the direct cause of the following exception:
    |
    | Traceback (most recent call last):
    | ConfigError: invalid 'port' setting

== add_note (3.11+) ==
   invalid literal for int() with base 10: 'x1' | notes: ['while reading row 7 of prices.csv']
```

(The script prints only the unindented traceback lines, to keep it short.)

## Common mistakes

- Wrapping errors *without* `from` and without a message — the new error hides what really happened.
- Using `from None` everywhere — you throw away debugging information. Use it only when the original is genuinely noise.

## In the interview

**30-second answer:** "Raising inside an `except` chains implicitly via `__context__`; `raise X from e` sets `__cause__` and says 'direct cause' in the traceback; `from None` hides the chain. Use explicit chaining when translating low-level errors into your own exception types. `add_note` adds context in 3.11+."

## Self-check

1. Attribute set by `raise A from b`? → `A.__cause__ = b`.
2. How do you hide the original exception? → `raise New(...) from None`.

## TL;DR

Translate errors with `raise MyError(...) from exc`; add context with `add_note`.

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## 9.5 EAFP vs LBYL

### What it is

Two styles for dealing with things that might not be there:

- **LBYL — Look Before You Leap**: check first, then act. `if key in d: v = d[key]`.
- **EAFP — Easier to Ask Forgiveness than Permission**: just try, and handle the exception. `try: v = d[key] except KeyError: ...`.

Both terms are in the official Python glossary, which calls EAFP the common Python style.

### Why it matters

EAFP avoids **race conditions** — the time-of-check-to-time-of-use (TOCTOU) gap. `if os.path.exists(p): open(p)` can still fail if another process (another running program) deletes the file between the check and the open; `try: open(p) except FileNotFoundError:` has no such gap. EAFP also avoids doing work twice (the check *and* the lookup) and plays well with duck typing (caring what an object can do, not what type it is — just call the method rather than checking the type).

```
stock = {"apple": 3}
qty = stock["pear"] if "pear" in stock else 0      # LBYL
try:                                               # EAFP
    qty = stock["pear"]
except KeyError:
    qty = 0
stock.get("pear", 0)                               # the idiomatic one-liner
```

```
  LBYL: 0
  EAFP: 0 | idiomatic: stock.get('pear', 0) = 0
```

### Trade-offs

| | EAFP | LBYL |
|---|---|---|
| Races (files, shared state) | Safe | Check can go stale |
| Speed when failure is **rare** | Fast (try is almost free when nothing is raised, 3.11+ "zero-cost") | Pays for the check every time |
| Speed when failure is **common** | Raising/catching is relatively expensive | Cheaper |
| Readability | Can hide which line failed if `try` is big | Explicit |

### When not to use it

Prefer LBYL when failure is the common case, when the "try" would have side effects you can't undo halfway, or when a dedicated API exists (`dict.get`, `getattr(obj, "x", default)`, `str.isdigit()` for simple inputs).

### In the interview

**30-second answer:** "LBYL checks before acting; EAFP attempts and catches. Python favours EAFP: it avoids race conditions like checking a file exists then opening it, and it's cheap when errors are rare. Use LBYL when failure is frequent or partial work is hard to undo."

### Self-check

1. Why is `if os.path.exists(p): open(p)` fragile? → The file can vanish between check and open.
2. Idiomatic "get with default" for dicts? → `d.get(key, default)`.

### TL;DR

Python prefers EAFP (try/except) — race-free and fast when errors are rare; use LBYL when failure is common.

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## 9.6 `ExceptionGroup` and `except*` (Python 3.11+)

### What it is

Sometimes **several** independent errors happen at once — concurrent tasks (tasks running at overlapping times), validating many fields, running many cleanup callbacks. An `ExceptionGroup` (PEP 654) wraps a list of exceptions into one raisable object. `except*` handles the matching *part* of a group: each `except*` clause receives a sub-group of the matching exceptions, and anything unhandled is re-raised in a new group.

### How it works

```
def run_all():
    errors = []
    for task, fn in [("a", lambda: 1 / 0), ("b", lambda: int("z")),
                     ("c", lambda: 42), ("d", lambda: [][1])]:
        try:
            fn()
        except Exception as exc:
            exc.add_note(f"task {task}")
            errors.append(exc)
    if errors:
        raise ExceptionGroup("some tasks failed", errors)

try:
    run_all()
except* ZeroDivisionError as eg:
    print("  math errors: ", [(repr(e), e.__notes__) for e in eg.exceptions])
except* (ValueError, IndexError) as eg:
    print("  input errors:", [repr(e) for e in eg.exceptions])
```

```
== ExceptionGroup and except* (3.11+) ==
  math errors:  [("ZeroDivisionError('division by zero')", ['task a'])]
  input errors: ['ValueError("invalid literal for int() with base 10: \'z\'")', "IndexError('list
index out of range')"]
```

Unlike plain `except`, **more than one** `except*` clause can run for one group.

Rules: `except` and `except*` can't be mixed in the same `try`; `eg` is always a group (use `eg.exceptions`); `ExceptionGroup` holds `Exception` subclasses, `BaseExceptionGroup` can hold any.

### Real-world examples

- `asyncio.TaskGroup` (3.11+; `asyncio` is Python's library for asynchronous code, and a TaskGroup runs several tasks together) raises an `ExceptionGroup` when child tasks fail.
- The Trio async library pioneered this model and was a key motivation for PEP 654.

### When not to use it

Ordinary single errors — use regular `except`. Groups are for genuinely concurrent/independent failures.

### In the interview

> **30-second answer:** "An `ExceptionGroup` bundles several exceptions; `except*` lets each clause handle the matching subset, and unhandled ones are re-raised. It's used by `asyncio.TaskGroup` when multiple tasks fail."

### Self-check

1. Can two `except*` clauses both run for one group? → Yes.
2. Where would you meet these in the stdlib? → `asyncio.TaskGroup`.

### TL;DR

Many errors at once → `ExceptionGroup`; split-handle with `except*`.

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## 9.7 `assert`

### What it is

`assert condition, "message"` raises `AssertionError("message")` if `condition` is false. It documents and checks **internal assumptions** — things that should be impossible if your code is correct.

```
def mean(xs):
    assert len(xs) > 0, "mean() of empty list"
    return sum(xs) / len(xs)
```

```
  mean([2, 4]) = 3.0
  AssertionError: mean() of empty list (disabled entirely by python -O)
```

**The crucial caveat**

Running Python with **`-O`** (optimise) sets `__debug__` to `False` and **removes all `assert` statements**. So never use `assert` for input validation, security checks or anything that must happen in production (the live system real users depend on) — raise a real exception (`ValueError`) instead.

### Common mistakes

- `assert (x > 0, "msg")` — a non-empty tuple is always truthy, so this never fails (CPython warns: "assertion is always true").
- Validating user input or permissions with `assert`.

### Real-world examples

**pytest** is built on plain `assert` statements — it rewrites them to show detailed failure messages (`assert result == 5` → shows both sides).

### In the interview

> **30-second answer:** "`assert` checks internal invariants (conditions that should always hold) and raises `AssertionError`; it's stripped under `python -O`, so use it for debugging and tests, never

for validating input."

### Self-check

1. What does `python -O` do to asserts? → Removes them.
2. What should validate a function argument from a user? → An explicit `raise ValueError(...)`.

### TL;DR

`assert` = developer sanity check and test tool; not validation.

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## 9.8 Context managers and `with`

### What it is

A **context manager** is an object that sets something up on entering a `with` block and **guarantees** teardown (the clean-up step) on leaving it — normally, by `return`/`break`, or by exception. Files, locks (which let only one thread use a resource at a time), database transactions (groups of changes that succeed or fail together), temporary directories and timers are typical.

```
with open("data.txt", encoding="utf-8") as f:   # file opened
    data = f.read()
# file closed here, even if read() raised
```

### How it works — the protocol

A *protocol* here means the set of special methods an object must provide. `with EXPR as NAME: BODY` does roughly:

1. Evaluate `EXPR` to get the manager `m`.
2. Call `m.__enter__()`; bind its return value to `NAME`.
3. Run `BODY`.
4. Call `m.__exit__(exc_type, exc, tb)` — with `None, None, None` if the body succeeded, or the exception details if it failed.
5. If `__exit__` returns a **truthy** value, the exception is **suppressed**; otherwise it propagates.

It's `try/finally` packaged as a reusable object.

```
class Timer:
    def __enter__(self):
        self.start = time.perf_counter()
        print("  __enter__: timer started")
        return self                     # bound to the name after 'as'

    def __exit__(self, exc_type, exc, tb):
        self.elapsed = time.perf_counter() - self.start
        print(f"  __exit__: exc_type={exc_type.__name__ if exc_type else None}")
        return False                    # let exceptions propagate
```

```
== class-based: normal exit ==
  __enter__: timer started
  __exit__: exc_type=None
  elapsed measured: True

== class-based: exception inside the block ==
  __enter__: timer started
  __exit__: exc_type=KeyError
  exception still reached caller: KeyError('boom')
```

`@contextlib.contextmanager` writes one as a generator (a function that pauses at `yield`): code before `yield` = enter, the yielded value = `as` target, code after = exit. Wrap the `yield` in `try` to react to errors:

```python
@contextlib.contextmanager
def transaction(log):
    log.append("BEGIN")
    try:
        yield log
    except Exception:
        log.append("ROLLBACK")
        raise
    else:
        log.append("COMMIT")
```

```
log: ['BEGIN', 'UPDATE accounts ...', 'COMMIT', 'BEGIN', 'UPDATE bad ...', 'ROLLBACK']
```

Other `contextlib` helpers used in the script:

| Helper | Does | Output |
|---|---|---|
| `suppress(FileNotFoundError)` | Ignore listed exceptions in the block | `missing file ignored` |
| `closing(obj)` | Call `obj.close()` on exit | |
| `ExitStack()` | Manage a *dynamic number* of managers; exits them in **reverse** order | `open db, cache, queue` → `close queue, cache, db` |
| Parenthesised `with (A() as a, B() as b):` | Several managers in one statement (3.10+) | |

## Trade-offs

Class-based managers are best when the object has other methods too (a connection). `@contextmanager` is shortest for one-off setup/teardown. `ExitStack` when the number of resources is only known at runtime.

## Common mistakes

- Opening files without `with` and forgetting `close()` — leaks file handles (the OS resources behind open files), and buffered writes (data held in memory before it is written out) may not reach disk.
- In a `@contextmanager` generator, forgetting `try/finally` around `yield` — the cleanup won't run if the body raises.
- Returning `True` from `__exit__` by accident → silently swallows every exception.

### Real-world examples

- `threading.Lock()`: `with lock:` acquires and always releases.
- `sqlite3` connection: `with conn:` commits (makes the changes permanent) on success and rolls back (undoes them) on exception (it does **not** close the connection).
- PyTorch `with torch.no_grad():`, `unittest.mock.patch(...)`, `decimal.localcontext()`, `tempfile.TemporaryDirectory()`.

Decorators and context managers get a deeper treatment (async managers, reentrancy — entering the same manager again while it is already active) in Chapter 12.

### In the interview

> **30-second answer:** "A context manager implements `__enter__` and `__exit__`; `with` calls enter, runs the block, and always calls exit with any exception info — exit returning true suppresses it. It's a reusable try/finally for resources like files and locks. `contextlib.contextmanager` builds one from a generator; `ExitStack` handles a dynamic set."

- *What does `__exit__` receive?* → `exc_type, exc_value, traceback`, all `None` on success.
- *How do you suppress an exception in a manager?* → Return `True` from `__exit__` (or use `contextlib.suppress`).

### Self-check

1. What is bound to `f` in `with open(p) as f`? → The return value of `__enter__` (the file object itself).
2. Order of exits in an `ExitStack`? → Reverse of entry (LIFO).
3. Does `with sqlite3.connect(...) as conn` close the connection? → No, it commits/rolls back.

### TL;DR

`with` = guaranteed cleanup. `__enter__`/`__exit__`, or `@contextmanager` with `try` around `yield`. `suppress`, `closing`, `ExitStack` from `contextlib`.

---

## 9.9 File I/O — modes, text vs binary, encodings

### What it is

`open(path, mode="r", encoding=None, newline=None)` returns a **file object**. In **text mode** it gives and takes `str`, decoding/encoding bytes with an **encoding** (a rule mapping characters to bytes, such as UTF-8). In **binary mode** (`"b"`) it gives and takes raw `bytes`.

| Mode | Meaning | If file exists | If missing |
|---|---|---|---|
| `"r"` (default) | read | read from start | `FileNotFoundError` |
| `"w"` | write | **truncates** (cuts) to empty | creates |
| `"a"` | append | writes go to the end | creates |
| `"x"` | exclusive create | `FileExistsError` | creates |
| `"+"` added (`"r+"`, `"w+"`) | read and write | | |
| `"b"` added (`"rb"`, `"wb"`) | binary (bytes) | | |
| `"t"` (default) | text (str) | | |

## How it works

```
notes = base / "notes.txt"
with open(notes, "w", encoding="utf-8") as f:    # 'w' creates/truncates
    f.write("first line\n")
    f.writelines(["second line\n", "café ☕\n"])
with open(notes, "a", encoding="utf-8") as f:    # 'a' appends
    print("appended line", file=f)
with open(notes, encoding="utf-8") as f:         # default mode 'r'
    for i, line in enumerate(f, 1):              # lazy, line by line
        print(f"  {i}: {line.rstrip()}")
```

```
== writing and reading text ==
  1: first line
  2: second line
  3: café ☕
  4: appended line
  read() whole file -> 44 characters

== text vs bytes, encodings ==
  bytes on disk: 47 (é is 2 bytes, ☕ is 3 bytes in UTF-8)
  first 12 bytes: b'first line\ns'
  ascii decode -> UnicodeDecodeError ordinal not in range(128) at byte 26
  latin-1 (wrong codec) mangles: 'cafÃ© â\x98\x95'
  errors='replace': caf�� ���

== binary mode, seek/tell ==
  tell after seek(4): 4 | read(3): [4, 5, 6]

== 'x' mode refuses to overwrite ==
  FileExistsError: File exists
```

Reading methods:

| Method | Returns | Memory |
|---|---|---|
| `for line in f:` | one line at a time (with `\n`) | Constant — best for big files |
| `f.read()` / `f.read(n)` | whole file / up to `n` chars (bytes in binary) | Whole file |
| `f.readline()` | next line | One line |
| `f.readlines()` | list of all lines | Whole file |
| `f.write(s)` | number of characters written | — |
| `f.seek(pos)`, `f.tell()` | move / report position | — |

Key facts:

1. **Always pass** `encoding="utf-8"` in text mode. Without it Python uses the platform's locale encoding (the default picked by the operating system's language/region settings), which differs between machines (historically cp1252 on many Windows systems) — the file you wrote on macOS may not read on Windows. PEP 686 plans to make UTF-8 the default in Python 3.15; until then, be explicit (or run with `-X utf8`).
2. The wrong codec either fails (`UnicodeDecodeError`) or, worse, silently produces **mojibake** (garbled text from decoding with the wrong encoding: `cafÃ©`). `errors="replace"` / `"ignore"` trade errors for data loss — last resorts.
3. Text mode translates newlines: on reading, `\r\n` and `\r` become `\n` ("universal newlines"); on writing, `\n` becomes the platform's separator. Binary mode never changes bytes.

4. Writes are **buffered**; data is guaranteed on disk after `close()` (or `flush()` + `os.fsync()` for durability — surviving a crash or power cut).
5. Images, zip files, pickles and anything non-text → binary mode.
6. **File-like objects**: anything with `read()`/`write()` can stand in for a file. `io.StringIO("text")` is an in-memory text file and `io.BytesIO(b"...")` an in-memory binary file — handy in tests and for libraries that expect a file (`csv.reader(io.StringIO(data))`, `pd.read_csv(io.StringIO(...))`).

### Common mistakes

- Opening with `"w"` when you meant `"a"` — the file is wiped instantly.
- Omitting `encoding=`.
- `f.read()` on a multi-GB log file — iterate lines instead.
- Building paths with string `+` (`folder + "/" + name`) — use `pathlib`.

### Real-world examples

Log processors stream files line by line; `pandas.read_csv(..., encoding="utf-8")` exposes the same encoding choice; almost every "garbled characters" bug report in data pipelines (chains of automated data-processing steps) is an encoding mismatch.

### In the interview

> **30-second answer:** "`open` returns a file object; use it in a `with` block. Text mode works with `str` and needs an encoding — always pass `utf-8` explicitly; binary mode works with `bytes`. `w` truncates, `a` appends, `x` fails if the file exists. Iterate over the file for large inputs."

- *Difference between `"w"` and `"x"`?* → `w` truncates an existing file; `x` raises `FileExistsError`.
- *Why can 'é' be 2 bytes?* → UTF-8 is variable-width: 1 byte for ASCII, 2–4 for other characters.

### Self-check

1. Mode for appending bytes? → `"ab"`.
2. What exception for decoding UTF-8 bytes as ASCII? → `UnicodeDecodeError`.
3. Memory-friendly way to process a huge text file? → `for line in f:`.

### TL;DR

`with open(p, mode, encoding="utf-8")`. Text = `str` + encoding, binary = `bytes`. `w` truncates, `a` appends, `x` is safe-create. Stream big files line by line.

---

## 9.10 `pathlib`, `shutil`, `tempfile`

### What it is

- **`pathlib.Path`** — an object-oriented path (a path stored as an object with its own methods). Join with `/`, inspect parts, and read/write/list without juggling strings or `os.path` functions. Works across Windows and POSIX (the Unix standard that Linux and macOS follow).
- **`shutil`** ("shell utilities") — high-level operations on files and whole trees: copy, move, delete trees, make archives.
- **`tempfile`** — safely creates temporary files and directories with unique names, cleaned up automatically.

## How it works — `pathlib`

```
p = base / "reports" / "2026" / "q3.final.csv"
p.name, p.stem, p.suffix, p.suffixes
p.parent.mkdir(parents=True, exist_ok=True)
p.write_text("id,total\n1,9.5\n", encoding="utf-8")
(base / "reports").glob("*/*.csv")
base.rglob("*.md")
```

```
== pathlib ==
  name: q3.final.csv | stem: q3.final | suffix: .csv | suffixes: ['.final', '.csv']
  parent relative to base: reports/2026
  with_suffix('.json'): q3.final.json
  exists: True | is_file: True | size: 15 bytes
  glob('*/*.csv'):    ['q2.csv', 'q3.final.csv']
  rglob('*.md'):      ['readme.md']
  iterdir(reports):   ['2026', 'readme.md']
```

| Task | `pathlib` | older `os` / `os.path` |
|---|---|---|
| Join | `base / "a" / "b.txt"` | `os.path.join(base, "a", "b.txt")` |
| Current dir / home | `Path.cwd()`, `Path.home()` | `os.getcwd()`, `os.path.expanduser("~")` |
| Absolute, symlinks (shortcut files pointing to another path) resolved | `p.resolve()` | `os.path.realpath(p)` |
| Parts | `p.name`, `.stem`, `.suffix`, `.parent` | `os.path.basename`, `splitext`, `dirname` |
| Exists / type | `p.exists()`, `.is_file()`, `.is_dir()` | `os.path.exists` … |
| Read / write whole file | `p.read_text()`, `p.write_bytes(b)` | `open(...)` |
| Make dirs | `p.mkdir(parents=True, exist_ok=True)` | `os.makedirs(p, exist_ok=True)` |
| Delete file / empty dir | `p.unlink(missing_ok=True)`, `p.rmdir()` | `os.remove`, `os.rmdir` |
| Rename / replace | `p.rename(t)`, `p.replace(t)` | `os.rename`, `os.replace` |
| List / search (glob = match names with wildcards like `*`) | `p.iterdir()`, `p.glob("*.py")`, `p.rglob("*.py")`, `p.walk()` (3.12+) | `os.listdir`, `glob.glob`, `os.walk` |

`open()` and nearly every library accept `Path` objects directly.

## How it works — `shutil` and `tempfile`

```
== shutil: copy, move, archive, delete trees ==
  copytree -> ['2026', '2026/q2.csv', '2026/q3.final.csv', 'readme.md']
  moved readme -> True
  archive: reports_bak.zip True
  rmtree backup -> exists? False
  disk_usage works: True

== tempfile ==
  temp file .txt contains: scratch
  deleted after close: True

== atomic replace pattern ==
  settings now: {"v": 2}

temp dir removed: True
```

| `shutil` function | Does |
|---|---|
| `copy(src, dst)` / `copy2` | Copy a file (+ permission bits — who may read/write/run it / + timestamps and other metadata, i.e. information about the file) |
| `copytree(src, dst)` | Copy a whole folder tree |
| `move(src, dst)` | Move/rename, across filesystems (separate disks or partitions) if needed |
| `rmtree(path)` | Delete a folder **and everything in it** — permanent, no recycle bin |
| `make_archive(base, "zip", root_dir)` / `unpack_archive` | Create/extract zip, tar, gztar… |
| `which("git")`, `disk_usage(path)` | Find an executable on `PATH`; total/used/free bytes |

| `tempfile` API | Gives |
|---|---|
| `TemporaryDirectory()` | A folder deleted (with contents) when the `with` block ends |
| `NamedTemporaryFile()` | A file with a real name, deleted on close by default (3.12 adds `delete_on_close=False` for reopening it by name, needed on Windows) |
| `TemporaryFile()` | Anonymous file — may have no visible name at all |
| `mkstemp()` / `mkdtemp()` | Low-level: create and return path; **you** delete it |

**Atomic replace pattern**: write the new content to a temp file *in the same directory*, then `os.replace(tmp, target)`. The rename is atomic (happens all at once or not at all) on the same filesystem, so readers see either the old or the new file — never a half-written one. Tools that save config and state files commonly use this.

## Trade-offs

`pathlib` is the modern default; `os`/`os.path` remain for low-level calls (`os.replace`, `os.fsync`, file descriptors — the OS's integer handles for open files) and older code. `shutil` operations are convenient but not transactional (all-or-nothing) — a `copytree` interrupted halfway leaves a partial copy.

### Common mistakes

- Hard-coding `"/tmp/myfile"` — predictable names are a security risk and not portable; use `tempfile`.
- `shutil.rmtree` on a path built from user input without checking it.
- Using `str` concatenation or backslashes for paths.
- `Path("a").glob("**/*.py")` vs `rglob("*.py")` — equivalent; `rglob` is shorter.

### Real-world examples

pytest's `tmp_path` fixture (a helper object pytest passes into tests) hands each test a fresh `pathlib.Path` temp directory. Build tools use `shutil.copytree`/`make_archive` for packaging; `shutil.which` is how scripts check that `git` or `ffmpeg` is installed.

### In the interview

> **30-second answer:** "`pathlib.Path` is the object-oriented way to build and inspect paths and to read, write, list and glob; `shutil` does high-level file and tree operations like copy, move, rmtree and archives; `tempfile` creates unique temp files and directories that clean themselves up. For safe file updates, write a temp file in the same folder and `os.replace` it."

### Self-check

1. Create `a/b/c` even if `a` doesn't exist and don't fail if `c` does? → `Path("a/b/c").mkdir(parents=True, exist_ok=True)`.
2. `Path("x.tar.gz").suffix`? → `".gz"` (`.suffixes` → `['.tar', '.gz']`).
3. Why write to a temp file then `os.replace`? → Atomic update; no half-written file.

### TL;DR

`Path` for paths, `shutil` for copying/moving/deleting trees, `tempfile` for throwaway files; atomic writes via temp + `os.replace`.

---

## 9.11 `csv`, `json`, `pickle`

### What it is

Three standard ways to store structured data in files:

| Format | Module | Looks like | Holds | Readable by other languages | Safe to load untrusted data |
|---|---|---|---|---|---|
| **CSV** — comma-separated values | `csv` | Text table | Rows of **strings** | Yes (Excel, databases) | Yes |
| **JSON** — JavaScript Object Notation | `json` | Text | dict, list, str, int, float, bool, None | Yes (the web's lingua franca) | Yes |
| **pickle** | `pickle` | Binary | Almost any Python object | **No** — Python only | **NO** — can run arbitrary (any, attacker-chosen) code |

#### CSV

Use the `csv` module, not `line.split(",")` — real CSV has quoted fields containing commas, quotes and even newlines.

```
with open(path, "w", newline="", encoding="utf-8") as f:   # newline="" per csv docs
    w = csv.DictWriter(f, fieldnames=["id", "customer", "total", "placed"])
    w.writeheader()
    for o in orders:
        w.writerow(asdict(o))
with open(path, newline="", encoding="utf-8") as f:
    rows = list(csv.DictReader(f))
```

```
  raw file:
    id,customer,total,placed
    1,"Ana, Inc.",19.5,2026-09-01
    2,"Bob ""B"" Ltd",5.25,2026-09-02
  first row: {'id': '1', 'customer': 'Ana, Inc.', 'total': '19.5', 'placed': '2026-09-01'}
  total is a str -> convert yourself: 24.75
  naive split(',') breaks on quoted commas: ['1', '"Ana', ' Inc."', '19.5', '2026-09-01']
  tab-separated: [['a', 'b'], ['1', '2']]
```

- The writer quoted `"Ana, Inc."` and doubled the inner quotes in `"Bob ""B"" Ltd"` automatically.
- **Everything reads back as `str`** — convert types yourself.
- Open with **`newline=""`** (the docs require it) so the `csv` module handles line endings itself; otherwise you can get blank lines between rows on Windows or broken quoted newlines.
- `csv.reader`/`csv.writer` work with lists; `DictReader`/`DictWriter` with dicts keyed by the header; `delimiter="\t"` for TSV (tab-separated values); `csv.Sniffer` guesses a dialect (the file's CSV variant: delimiter, quoting rules).
- For analysis-sized data, `pandas.read_csv` is usually the better tool.

### JSON

| Python → JSON | JSON → Python |
|---|---|
| `dict` → object | object → `dict` |
| `list`, `tuple` → array | array → `list` |
| `str` → string | string → `str` |
| `int`, `float` → number | number → `int` or `float` |
| `True`/`False` → `true`/`false` | → `True`/`False` |
| `None` → `null` | → `None` |

`dumps`/`loads` work with strings; `dump`/`load` with files.

```
text = json.dumps(payload, default=to_json_default, indent=2, ensure_ascii=False)
back = json.loads(text)
```

```
  first lines of dumps():
    {
      "orders": [
        {
          "id": 1,
          "customer": "Ana, Inc.",
          "total": 19.5,
          "placed": "2026-09-01"
        },
  loads() types: paid -> bool | discount -> None | placed -> str
  load() from file, customers: ['Ana, Inc.', 'Bob "B" Ltd']
  tuple becomes list: {'pt': [1, 2]}
  int keys become str: {'1': 'a'}
  without default=: Object of type date is not JSON serializable
  JSONDecodeError: Expecting property name enclosed in double quotes at pos 1
```

- Types JSON doesn't know (`date`, `Decimal`, `set`, your classes) need a `default=` function (or pre-conversion, e.g. `dataclasses.asdict`); they come back as plain strings/lists — the round trip is **lossy** (information is lost) for types.
- Tuples become lists; non-string keys become strings.
- JSON requires double quotes; `'single'` quotes (a Python repr) are invalid → `json.JSONDecodeError` (a `ValueError` subclass).
- `indent=2` pretty-prints; `ensure_ascii=False` keeps "é" readable instead of the escape `\u00e9`; `sort_keys=True` gives stable output. `python -m json.tool file.json` pretty-prints from the shell.
- By default `json.dumps(float("nan"))` writes `NaN`, which is **not** valid JSON for strict parsers; pass `allow_nan=False` to get an error instead.

### pickle

`pickle` serialises (turns into bytes) almost any Python object graph (an object plus everything it references) (custom classes, dates, nested references) to bytes and back.

```
blob = pickle.dumps(orders)
restored = pickle.loads(blob)
with open(base / "orders.pkl", "wb") as f:                # binary mode!
    pickle.dump(orders, f)
```

```
  round trip equal: True | type: Order | placed: 2026-09-01
  load() from file: 2 orders
```

**Security caveat — never unpickle data you don't fully trust.** The pickle format lets an object specify, via `__reduce__`, a function to *call* when it's loaded. The script demonstrates with a harmless `print`; an attacker would use `os.system` (runs any shell command) or similar:

```
class Sneaky:
    def __reduce__(self):
        return (print, ("  !!! code ran during pickle.loads() !!!",))

pickle.loads(pickle.dumps(Sneaky()))
```

```
  loading a crafted pickle...
  !!! code ran during pickle.loads() !!!
  loads() returned: None
```

The code ran merely by *loading*. The official docs carry a prominent warning to this effect.

Other pickle limits: Python-only; tied to your class definitions (rename a class and old pickles break); lambdas, open files, sockets and locks can't be pickled. The current protocol is 5 (`pickle.HIGHEST_PROTOCOL`).

## Trade-offs / when to use which

- **Config, APIs, data exchange** → JSON (or TOML for human-edited config, which Python reads with `tomllib`).
- **Tabular data for spreadsheets/other tools** → CSV (or Parquet — a compressed, column-oriented data file format — via pandas for big data).
- **Short-lived caching of Python objects you created yourself** (e.g. between steps of your own pipeline, `multiprocessing` internals) → pickle.
- **Never** pickle for data crossing a trust boundary (the line between data you control and data from outside) (uploads, network, downloaded files).

## Common mistakes

- Opening pickle files in text mode (`"w"`) → `TypeError`; always `"wb"`/`"rb"`.
- Forgetting `newline=""` with `csv`.
- Expecting `json.loads` to give back `datetime`s.
- Loading downloaded model files with plain `pickle.load`.

## Real-world examples

- **PyTorch** checkpoints (saved snapshots of a model; `torch.save`) use pickle; since PyTorch 2.6, `torch.load` defaults to `weights_only=True` to block arbitrary code during loading, and Hugging Face created the **safetensors** format as a pickle-free way to share model weights (the learned numbers inside an ML model).
- scikit-learn documents pickle/joblib for model persistence (saving to disk to reload later) with the same security warning, and points to the `skops` format as a safer option.
- REST APIs (web APIs built from plain HTTP requests to URLs; GitHub, Stripe) speak JSON; spreadsheet exports are CSV.

## In the interview

> **30-second answer:** "CSV is text rows of strings — use the `csv` module with `newline=''` because of quoting. JSON maps to dict/list/str/number/bool/None, is language-neutral and safe; other types need a `default=` hook and come back as plain values. Pickle stores arbitrary Python objects in binary but is Python-only and unsafe: loading can execute code via `__reduce__`, so never unpickle untrusted data."

- *Why is `pickle.loads` dangerous?* → A pickle can instruct the loader to call arbitrary callables.
- *How do you JSON-encode a `datetime`?* → `default=` function returning `isoformat()`, or a custom `JSONEncoder`.
- *`json.dump` vs `json.dumps`?* → File vs string.

## Self-check

1. What type are all values from `csv.DictReader`? → `str`.
2. What does `json.loads(json.dumps((1, 2)))` return? → `[1, 2]`.
3. Mode to write a pickle? → `"wb"`.
4. Safe alternative to pickle for sharing ML weights? → safetensors (or `torch.load(..., weights_only=True)`).

**TL;DR**

CSV = strings in rows (`csv` module, `newline=""`). JSON = safe, portable, limited types. Pickle = any Python object, binary, **never from untrusted sources**.

---

## Chapter TL;DR

- Exceptions are objects that unwind the stack until an `except` matches by class; `Exception` is the base for application errors, `BaseException` also covers exit and Ctrl-C.
- `try` small, `except` specific (first match wins), `else` on success, `finally` always; never a bare `except:`; the `as e` name is deleted after the handler.
- `raise` built-ins for contract errors, a custom hierarchy for domain errors, bare `raise` to re-throw, `raise X from e` to translate while keeping the cause, `add_note` for context.
- Python prefers **EAFP** (try it) over LBYL (check first) — race-free and fast when failures are rare.
- `ExceptionGroup` + `except*` handle several simultaneous errors (e.g. `asyncio.TaskGroup`).
- `assert` checks invariants and vanishes under `-O` — never for validation.
- `with` + context managers guarantee cleanup (`__enter__`/`__exit__`, `@contextmanager`, `ExitStack`).
- `open(path, mode, encoding="utf-8")` inside `with`; text vs binary; `w` truncates, `a` appends, `x` refuses to overwrite; iterate big files.
- `pathlib` for paths, `shutil` for trees and archives, `tempfile` for throwaway files, temp + `os.replace` for atomic writes.
- CSV (strings, `newline=""`), JSON (safe, portable, limited types), pickle (any object, Python-only, **unsafe on untrusted data**).